SOC-5811 Week 6: Simulation

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Review

- Sampling distributions

Review

- Sampling distributions
- Statistical inference and p-values

Review

- **Standard deviation** = a measure of the spread of data around the sample mean.

$$\sigma = \sqrt{\frac{\sum(x_i - \mu)^2}{n}}$$

- **Standard error** = a measure of the spread of sample means around the population mean; the standard deviation of the sampling distribution.

$$\text{Standard error} = \frac{\sigma}{\sqrt{n}}$$

SIMULATION

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- This week is **simulation**: We *know* the population model $\longrightarrow$ we simulate all the sample data is could plausibly produce (i.e., we simulate the *sampling distribution*).

Simulation

Why is simulation an important concept/skill?

- Develop statistical intuition without needing math.

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Why is simulation an important concept/skill?

- Develop statistical intuition without needing math.
- We can use simulation to approximate the sampling distribution of data and propagate this to the sampling distribution of statistical estimates (aka **bootstrapping**).
- Very important principle for more complex models: you can always simulate.

Simulation

- Remember: this is all about understanding how we do **statistical inference**.

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- What can I learn about the population model from the data I observe?
- Are my sample parameters real, or could they plausibly have been observed by chance?

SIMULATION

Table 2. Children with Same-Sex Parents and Standardized Test Scores at the End of Primary Education

	All Children		Children Raised by Same-Sex Parents from Birth[a]		
	(1)	(2)	(3)	(4)	(5)
Child has same-sex parents (1 is yes)	.106***	.054**	.194***	.139***	.147***
	(.019)	(.018)	(.024)	(.023)	(.041)
Sex (1 is male)	−.004*	−.009***	−.003	−.007***	
	(.002)	(.002)	(.002)	(.002)	
Ethnicity (ref: both parents born in NL)					
One parent NL, other Western country		.004		.004	
		(.003)		(.003)	
One parent NL, other non-Western country		−.031***		−.031***	
		(.005)		(.005)	
Both parents not NL		−.111***		−.111***	
		(.006)		(.006)	
Parental education (1 is diploma SE)		.531***		.532***	
		(.004)		(.004)	
Log household income		.083***		.084***	
		(.002)		(.002)	
Mean parental age		.018***		.018***	
		(.000)		(.000)	
Family size		−.066***		−.066***	
		(.001)		(.001)	
Family structure (ref: married parents)					
Cohabiting parents		−.073***		−.073***	
		(.002)		(.002)	
Other		−.367***		−.154***	
		(.012)		(.032)	
Fixed effects					
Birth year	Yes	Yes	Yes	Yes	
Neighborhood	No	Yes	No	Yes	
Method	OLS	OLS	OLS	OLS	CEM[b]
Number of children	1,204,692	1,204,692	1,195,624	1,195,624	48,388
Number of children with same-sex parents	2,971	2,971	1,390	1,390	1,390
R^2	.004	.108	.004	.109	.121

Simulation

Table 1

Estimates of the Effect of Veteran Status on Homeownership by Race (1960)

	Outcome	All	White	Black
Stage 1	Veteran status	.396***	.409***	.274***
		(.005)	(.005)	(.009)
Reduced Form	Homeownership	.048***	.051***	.020**
		(.003)	(.003)	(.005)
IV	Homeownership	.121***	.125***	.071***
		(.008)	(.009)	(.019)
OLS	Homeownership	.114***	.104***	.080***
		(.006)	(.007)	(.011)

This week: R skills

- Functions in R

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- Simulating in R

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- Simulating in R
- Bootstrapping in R

This week: R skills

- Functions in R
- Simulating in R
- Bootstrapping in R
- Introduction to multivariable models

Functions in R

```r
log(5)
```

```
## [1] 1.609438
```

Functions in R

```
new_function <- function(x) {
  y <- x*10
  return(y)
}
```

FUNCTIONS IN R

```r
new_function <- function(x) {
  y <- x*10
  return(y)
}
new_function(5)
```

```
## [1] 50
```

FUNCTIONS IN R

```
new_function <- function(x) {
  y <- x*10
  return(y)
}
sapply(c(5,10,15,20), new_function)
```

```
## [1]  50 100 150 200
```

SIMULATION

- Random variables

Simulation

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- Distributions:

Simulation

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SIMULATION

- Random variables
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- **Simulation of random variables in R.**

Simulation

X = the number of heads I get if I flip a coin 100 times.

X = Binomial(n=100, p=0.5)

```
rbinom(n=1, size=100, prob=0.5)
```

```
## [1] 52
```

Simulation

```
rbinom(n=1, size=100, prob=0.5)
```

```
## [1] 51
```

```
rbinom(n=1, size=100, prob=0.5)
```

```
## [1] 46
```

```
rbinom(n=1, size=100, prob=0.5)
```

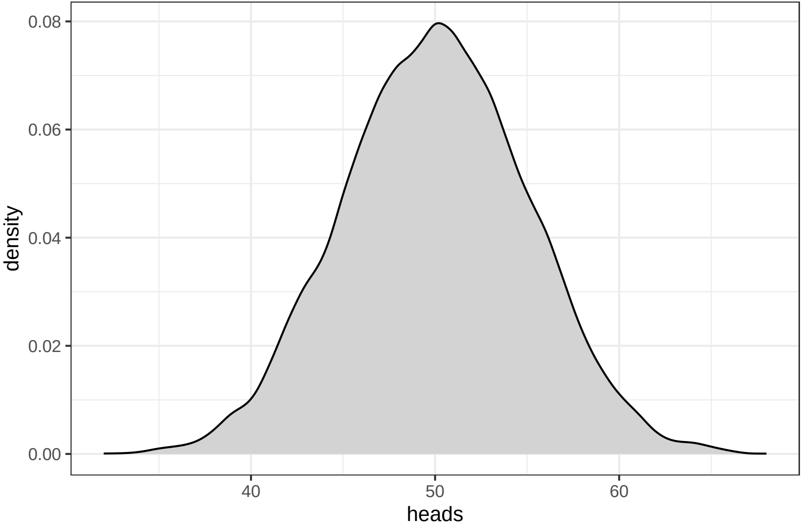
```
## [1] 56
```

Simulation

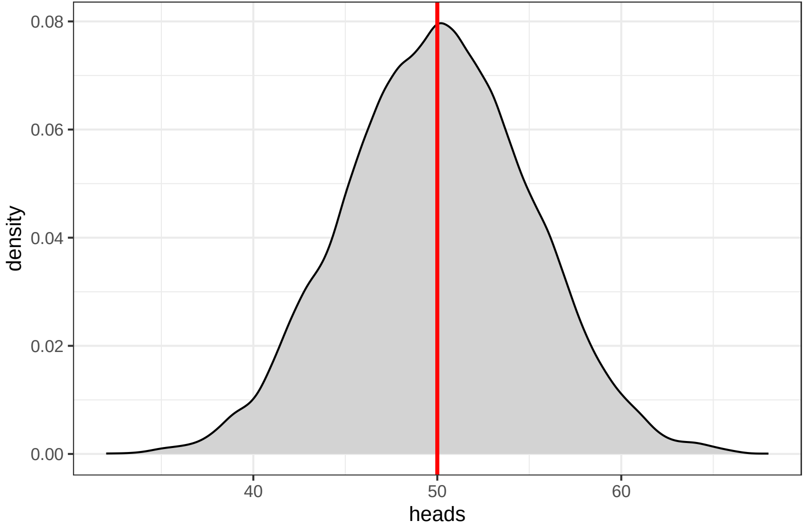
```r
trials <- rbinom(n=10000, size=100, prob=0.5)
trials <- data.table(heads=trials)
head(trials)
```

```
##    heads
##    <int>
## 1:    60
## 2:    50
## 3:    42
## 4:    57
## 5:    50
## 6:    52
```

SIMULATION



Simulation



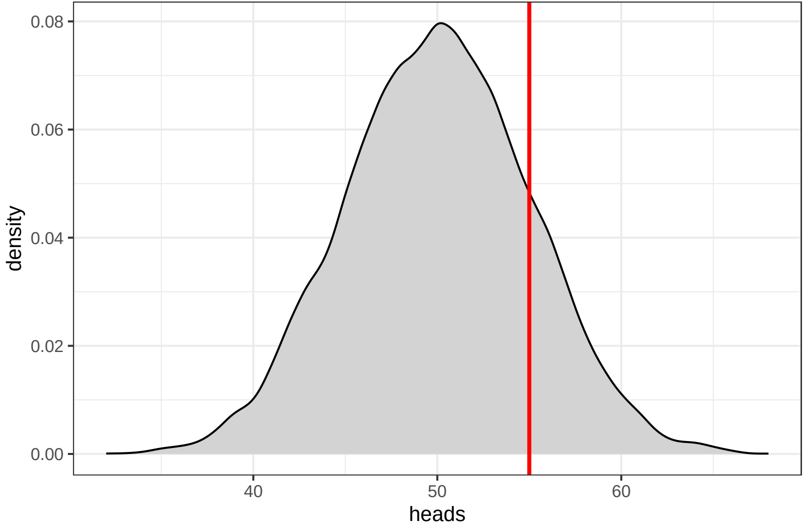
Simulation

Imagine flipping a coin 100 times and interpreting the results relative to some null probability model.

Null model: $P(\text{heads}) = 0.5$; this is a fair coin.

Results: 55 heads, 45 tails.

Simulation



Simulation

```r
pbinom(55, 100, 0.5, lower.tail = FALSE)
```

```
## [1] 0.1356265
```

Simulation

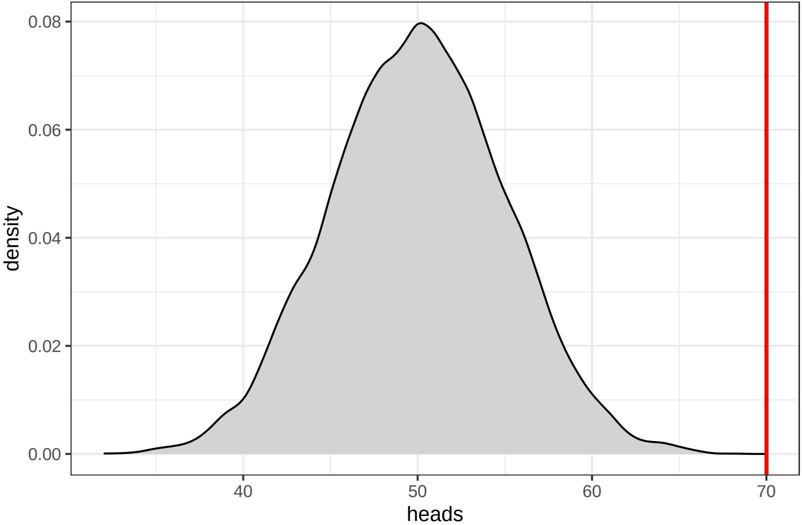
Imagine flipping a coin 100 times and interpreting the results relative to some null probability model.

Null model: $P(\text{heads}) = 0.5$; this is a fair coin.

Results: 55 heads, 45 tails.

Is the hypothesis that this is a fair coin consistent with our data? Could a fair coin plausibly produce our data?

Simulation



Simulation

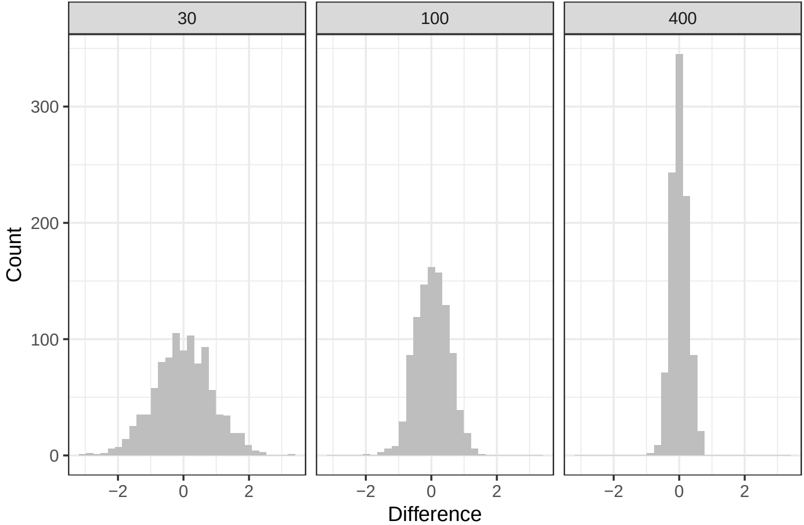
```
pbinom(70, 100, 0.5, lower.tail = FALSE)
```

```
## [1] 1.608001e-05
```

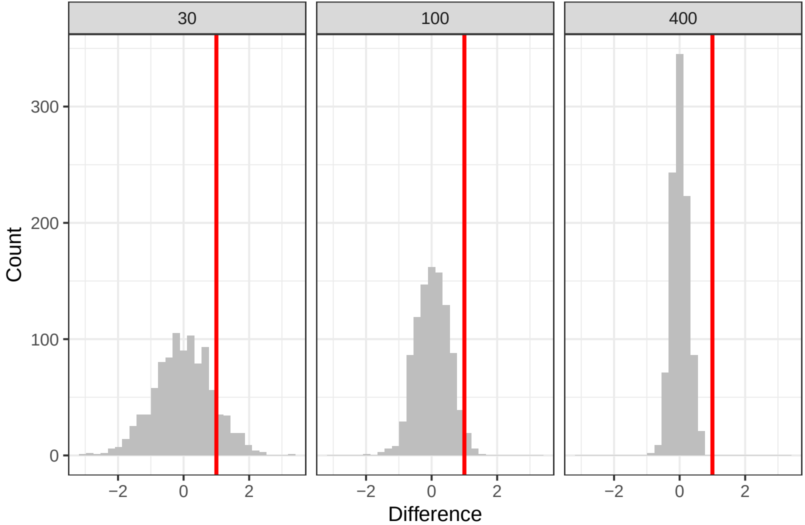
Simulation

```
mean_of_normal_trials <- function(i, ss) {
  mean <- mean(rnorm(ss, 0, 5))
  return(mean)
}
normal_trials <- function(ss) {
  trials <- lapply(1:1000,
                   mean_of_normal_trials,
                   ss=ss)
  trials <- data.table(sample_size=ss,
                       mean=unlist(trials))
  return(trials)
}
trials <- lapply(c(30,100,400), normal_trials)
trials <- rbindlist(trials)
```

Simulation



Simulation



Bootstrapping

- Simulating a sampling distribution when you **do not know** the underlying probability model.
- Randomly sample the **observed** data and calculate your statistic many times ("bootstrapping").

Bootstrap

```
psid <- fread('https://raw.githubusercontent.com/ngraetz/soc5811/refs/heads/main/data/psid.csv
psid <- psid %>%
    filter(g3_house_value!=0)
```

Bootstrap

```r
mean_analytic <- psid %>%
  summarise(mean=mean(g3_house_value)) %>%
  pull(mean)
se_analytic <- psid %>%
  summarise(se=sd(g3_house_value)/sqrt(nrow(psid))) %>%
  pull(se)
mean_analytic
```

```
## [1] 388711.5
```

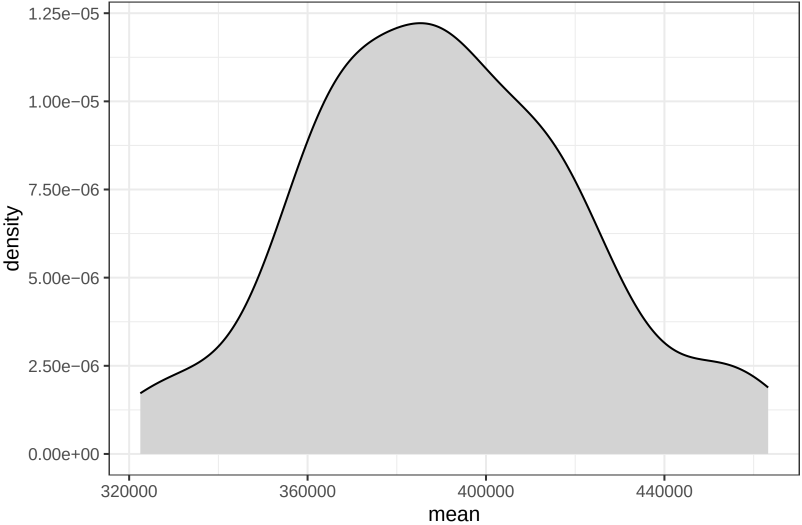
```r
se_analytic
```

```
## [1] 31072.29
```

Bootstrap

```r
get_sample <- function(i) {
  random_sample <- psid %>%
    sample_n(size=nrow(psid),replace=T) %>%
    summarise(mean=mean(g3_house_value))
  return(random_sample)
}
samples <- rbindlist(lapply(1:100,get_sample))
```

Bootstrap


density
1.25e−05
1.00e−05
7.50e−06
5.00e−06
2.50e−06
0.00e+00
320000
360000
400000
440000
mean

Bootstrap

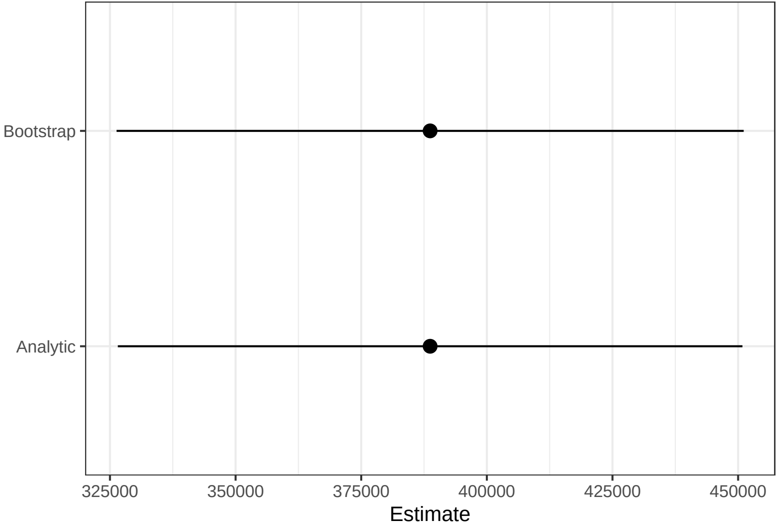
```r
## Calculate standard error
se_bootstrap <- samples %>%
  summarise(sd=sd(mean)) %>%
  pull(sd)

## Compare
se_analytic
```

```
## [1] 31072.29
```

```r
se_bootstrap
```

```
## [1] 31198.43
```

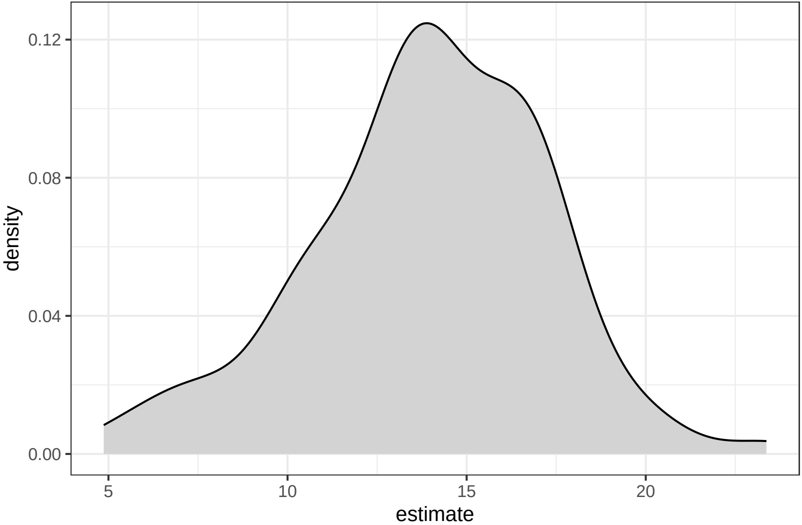
Bootstrap


Bootstrap
Analytic
325000
350000
375000
400000
425000
450000
Estimate

Bootstrap

```
psid_subset <- psid %>%
  filter(!is.na(g1_edu_cat) & g3_house_value!=0) %>%
  mutate(above_hs=ifelse(g1_edu_cat %in% c('less_hs','hs
                       'less', 'more'))

get_sample <- function(i) {
  random_sample <- psid_subset %>%
    sample_n(size=nrow(psid_subset),replace=T)
  model <- lm(g3_house_value_percentile~above_hs,
              data=random_sample)
  out <- avg_slopes(model)
  return(out)
}
coef_samples <- rbindlist(lapply(1:100,get_sample))
```

Bootstrap



Bootstrap

```r
## Compare to lm()
model <- lm(g3_house_value_percentile~above_hs,
            data=psid_subset)
analytic_coefs <- as.data.table(avg_slopes(model))
```

Bootstrap